

## Report R06 — Chromatin-Regulator Mutations and Survival in NSCLC

The reframed lung arm. One cohort, not two — the audit withdrew the replication claim.

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### Summary

The response letter proposed abandoning the acquired-resistance framing, which public data cannot power, in favour of chromatin dysregulation. This report tests that directly on mutation and survival data — no imaging, no methylation download required.

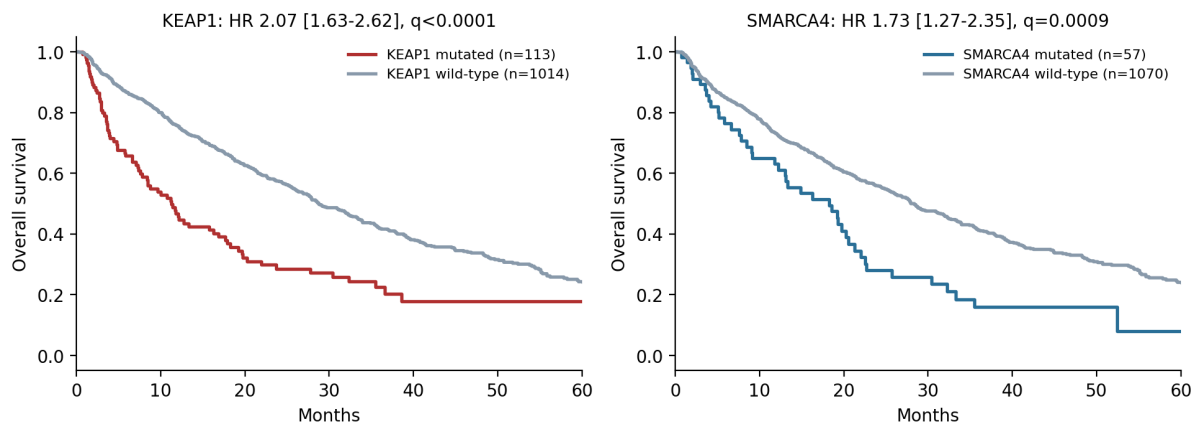
In the MSK NSCLC circulating-tumour-DNA cohort ( $n = 1,127$ , **618 events**), **KEAP1** (HR 2.07, 95% CI 1.63–2.62,  $q < 0.0001$ ) and **SMARCA4** (HR 1.73, 1.27–2.35,  $q = 0.0009$ ) mutations are both associated with shorter overall survival. Median survival falls from 28.8 to 11.3 months with KEAP1 mutation and from 28.4 to 18.2 months with SMARCA4. Both pre-registered positive controls were recovered at expected magnitudes (STK11 HR 1.63,  $q = 0.0004$ ; TP53 HR 1.25,  $q = 0.0074$ ).

**This is a single-cohort finding.** The run's decision line claimed SMARCA4 replicated across two cohorts; the audit withdrew that, for reasons in §3.

### 1 Results — MSK NSCLC ctDx cohort ( $n = 1,127$ )

| Gene                     | Mutated | HR          | 95% CI    | $q$ (BH)   | Median OS, mut vs WT |
|--------------------------|---------|-------------|-----------|------------|----------------------|
| <b>KEAP1</b>             | 113     | <b>2.07</b> | 1.63–2.62 | $< 0.0001$ | 11.3 vs 28.8 mo      |
| <b>SMARCA4</b>           | 57      | <b>1.73</b> | 1.27–2.35 | 0.0009     | 18.2 vs 28.4 mo      |
| KMT2D                    | 39      | 0.98        | 0.65–1.49 | 0.94       | 27.8 vs 27.4 mo      |
| STK11 ( <i>control</i> ) | 106     | 1.63        | 1.27–2.10 | 0.0004     | 14.0 vs 28.0 mo      |
| TP53 ( <i>control</i> )  | 580     | 1.25        | 1.07–1.46 | 0.0074     | 23.8 vs 32.3 mo      |

MSK NSCLC ctDx cohort,  $n=1127$ , 618 events



### 2 Why the replication claim was withdrawn

The second cohort (luad\_mskcc\_2020,  $n = 604$ ) appeared to corroborate SMARCA4 at HR 3.97 ( $q = 0.001$ ). It cannot be used for that purpose:

- It has only **45 events among 604 patients (7%)**.
- **It fails its own positive controls.** TP53 HR 1.18 ( $p = 0.58$ ) and STK11 HR 1.04 ( $p = 0.91$ ) — two established NSCLC prognostic factors, both undetected. A cohort that cannot find STK11 or TP53 cannot corroborate SMARCA4.
- Its SMARCA4 estimate rests on **9 events among 33 mutants**, with a confidence interval spanning 1.90–8.29.
- KEAP1 pointed in *opposite* directions between the two cohorts (HR 0.87 vs 2.07), which is the signature of an underpowered comparison.
- **KMT2D’s reported HR of 0.000 is a degenerate fit, not an estimate** — there are zero events among its 28 mutants, so the hazard ratio is not estimable. It should not have been printed as a number.

The honest statement is therefore single-cohort, well-controlled, and not yet replicated.

### 3 Interpretation

KEAP1 and SMARCA4 both plausibly belong to the same biology the reframed arm targets. KEAP1 loss deregulates the NRF2 oxidative-stress programme; SMARCA4 is a core SWI/SNF chromatin remodeller. That SMARCA4 is prognostic here is consistent with the SWI/SNF tyrosine-kinase-inhibitor-resistance literature that motivated the original lung manuscript — but as a prognostic association in a mostly treatment-heterogeneous cohort, not as evidence of a resistance mechanism.

KMT2D, the third chromatin regulator and the one with the largest patient count in our earlier inventory, is **null** (HR 0.98). The reframed arm should not be built on KMT2D.

### 4 Limitations

- **Single cohort.** Replication requires a second adequately powered cohort; neither of ours qualifies.
- **Targeted panel, circulating tumour DNA.** Mutation detection depends on tumour shedding, and ctDNA-positive disease is itself prognostically adverse. Effect sizes may partly reflect detectability.
- **No treatment adjustment.** The cohort spans heterogeneous therapy; no line of treatment or regimen was adjusted for.
- **No methylation.** The chromatin-*dysregulation* framing would be far stronger with the methylome. That download (TCGA-LUAD/LUSC, 1,283 open files) remains the main funded extension.
- Repository `data_mutations.txt` for both cohorts are 132-byte git-LFS pointer stubs whose objects are unretrievable upstream; all mutation data here came from the cBioPortal REST API instead. Recorded as gate G1.

### 5 Audit

| Check                           | Verdict                        | Finding                                                                |
|---------------------------------|--------------------------------|------------------------------------------------------------------------|
| Positive controls, large cohort | <b>Both recovered</b>          | STK11 $q = 0.0004$ ; TP53 $q = 0.0074$                                 |
| Positive controls, small cohort | <b>Both FAILED</b>             | TP53 $p = 0.58$ ; STK11 $p = 0.91$ — cohort unusable for corroboration |
| Event adequacy                  | <b>Small cohort inadequate</b> | 45 events / 604 patients; SMARCA4 rests on 9 events                    |
| Degenerate fits                 | <b>One found</b>               | KMT2D HR 0.000, zero events among 28 mutants; not estimable            |
| Direction consistency           | <b>KEAP1 inconsistent</b>      | HR 0.87 vs 2.07 across cohorts                                         |
| Gates                           | <b>7 / 7 PASS</b>              | Patient counts, API retrieval, survival counts, controls               |

**Note on the evidence bundle:** `results.json` contains a `_decision` field stating that SMARCA4 replicates. That line reflects the pre-specified rule as mechanically applied and is *superseded by this report*. The decision fields in run artefacts should not be cited without the accompanying audit.

## 6 How to verify

`evidence/results.json` holds every hazard ratio, confidence interval and  $q$ -value for both cohorts. `config.yaml` (hash `d3919b85...`) was written before any outcome was read. Mutation retrieval is reproducible with the API call in `m6_lung_chromatin.py`.